

SYED IRFAN M

Full Stack Developer | AI/ML Engineer | B.Tech AIML – SRMIST Trichy (2027)

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EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

2023 – 2027

SRMIST Trichy

CGPA: 8.00 / 10

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Machine Learning, Software Engineering

TECHNICAL SKILLS

Languages:	Java, Python, JavaScript (ES6+), TypeScript, C
Backend:	Spring Boot, Node.js, Express.js, RESTful API Design, Microservices, JWT Authentication
Frontend:	React.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design, Figma
Databases:	MySQL, SQLite, SQL Query Optimization, Indexing, ORM (Hibernate, Sequelize)
AI / ML:	TensorFlow, Keras, Scikit-learn, OpenCV, MediaPipe, CNN, LSTM, NLP, Random Forest, Whisper ASR
Tools:	Git, GitHub, Docker, Postman, npm, VS Code, Linux, Jupyter Notebook, Flask
Data & BI:	Tableau, Power BI, Excel, ETL, Data Visualization, GeoPandas
Concepts:	DSA, OOP, DBMS, OS, CN, System Design, Agile/Scrum, SDLC, RHCSA

WORK EXPERIENCE

Machine Learning Intern | APPROTECH

2025

- Developed and deployed end-to-end ML pipelines for real-world workflows using Python, Scikit-learn, and TensorFlow; maintained clean, well-documented code aligned with team coding standards.
- Built and integrated RESTful APIs using Flask to expose model inference endpoints consumed by downstream application services.
- Collaborated with cross-functional teams across feature delivery cycles; participated in code reviews and followed Git-based version control best practices throughout the development lifecycle.
- Optimized model preprocessing pipelines, reducing average prediction latency and improving overall application responsiveness for end users.

PROJECTS

Sign Language to Text Conversion | *Python, TensorFlow, OpenCV, MediaPipe, Random Forest* (2025–2026)

- Trained CNN and Random Forest classifier on custom gesture dataset achieving **92%+ accuracy**; integrated real-time webcam pipeline using TensorFlow and OpenCV for continuous inference.
- Extracted 21 hand landmark features per frame via MediaPipe enabling robust recognition across 26 sign classes with low-latency CPU-based prediction – no specialized hardware required.
- Published at ICTMIM International Conference (2026). GitHub: github.com/IrfanM-7/Sign-Language-to-Text-conversion

Network Intrusion Detection System | *Python, CNN, LSTM, TensorFlow, NSL-KDD* (2025)

- Built a deep learning model using **CNN-LSTM** architecture to detect cyber attacks (DoS, Probe, R2L) on the NSL-KDD dataset, learning spatial and temporal traffic patterns for enhanced detection accuracy.
- Applied SMOTE and ADASYN oversampling to address class imbalance, significantly boosting minority class F1-score over baseline models.
- Processed and engineered features from large-scale network traffic datasets; optimized performance through iterative experimentation. GitHub: github.com/IrfanM-7

Nagarvaani – Tamil AI Communication Ecosystem | *Python, Whisper ASR, NLP, Gradio* (2025)

- Built an AI-powered governance platform integrating Tamil chatbot, voicebot, and NLP-based civic complaint management system for regional government, reducing manual categorization effort significantly.
- Implemented speech-to-text with Whisper ASR and automated complaint classification pipeline for faster government response workflows.
- Designed responsive Gradio UI for real-time citizen-government interaction; presented at AICTE-VAANI, Modern Tamil Computing Summit (2025). GitHub: github.com/IrfanM-7/Nagarvaani

AgriConnect – Agricultural Supply Chain Platform | *React.js, Node.js, Express.js, Firebase (2025)*

- Built a full-stack real-time marketplace connecting farmers, buyers, and experts with intelligent multi-factor matching (crop type, price, urgency) using React.js, Node.js, and RESTful API architecture.
- Implemented normalized database schema with role-based access control and Firebase real-time synchronization across concurrent users with sub-second latency. GitHub: github.com/IrfanM-7/AgriConnect

RESEARCH & PUBLICATIONS

Nagarvaani: Tamil AI Communication Ecosystem | *AICTE-VAANI, Modern Tamil Computing Summit (2025)*

- Developed AI governance framework featuring Tamil Chatbot, Voicebot, and NLP-based Civic Complaint Management System using Whisper ASR for automated complaint classification.
- Designed responsive Gradio UI for seamless Tamil-language citizen-government interaction; presented at national-level AI summit.

Sign Language to Text Conversion | *ICTMIM International Conference (2026)*

- Engineered real-time system converting sign language alphabets and digits to text using MediaPipe hand landmarks and Random Forest Classifier for low-latency CPU-based prediction.

CERTIFICATIONS

- **RHCSA (Red Hat Certified System Administrator)** – Red Hat
- **Introduction to Large Language Models** – Google Cloud / Coursera
- **Intro to Generative AI** – Google Cloud / Coursera
- **AI Infrastructure and Operations Fundamentals** – NVIDIA / Coursera
- **Database Structures and Management with MySQL** – Meta / Coursera

HACKATHONS & ACHIEVEMENTS

YUVA Hackathon – AI & Language Tech Sector | **SRMIST**

Sep 2025

- Built a **Smart City Infrastructure Optimizer** using AI, geospatial analysis (GeoPandas), and LLMs (GPT-4/Gemini) with Flask APIs; generated data-driven cost-benefit reports with interactive dashboards.
- Visualized population density and traffic congestion from large geospatial datasets using scenario-based planning and automated recommendation pipelines.

BIT V-PRAYUKTI'25 – National Level AIML Hackathon

2025

- Developed a mood-aware conversational AI using NLP, sentiment analysis, and GPT-based dynamic question generation to adapt dialogue based on detected user emotional context for mental wellness support.